Author perfomance prediction

Laboratory 1, Social Media Analytics

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Predicting Post Performance

Goal: classify each LinkedIn post relative to its author’s usual performance.

Below (0)	performs worse than usual
Typical (1)	performs around the author’s usual level
Above (2)	performs better than usual
Primary metric	Macro-F1

Data and Evaluation

- 71,395 labelled posts from training authors
- 17,623 labelled posts from test authors
- Validation split: 56,847 fit posts and 14,548 validation posts
- Authors are disjoint between fit, validation, and test
- Model selection was based on validation Macro-F1
- Test metrics are reported for final evaluation
- Initial tabular baseline: 0.5410 Test Macro-F1

Feature Engineering: Post, Text, and Author History

- Post length, structure, punctuation, hashtags, links, and content type
- Posting time and author activity
- Text features compared with the author’s earlier posts
- MiniLM embeddings represent the meaning of each post
- Long posts are split into chunks so the full text is represented

Experiments

Model	Validation Macro-F1	Test Macro-F1
Initial tabular HistGradientBoosting baseline	--	0.5410
HistGradientBoosting + TF-IDF/SVD + engineered features	0.6039	0.5981
LightGBM + TF-IDF/SVD (200 components)	0.6291	0.6196
Regularized neural network	0.6321	--
Two-branch neural network	0.6340	--
Full-text MiniLM + LightGBM + author-history features	0.6406	0.6305

Conclusions

- The final LightGBM model achieved 0.6406 Validation Macro-F1 and 0.6305 Test Macro-F1.
- Test Macro-F1 improved from 0.5410 for the baseline to 0.6305
- Media was not used due to limited computing resources. The model relied on post text and tabular features